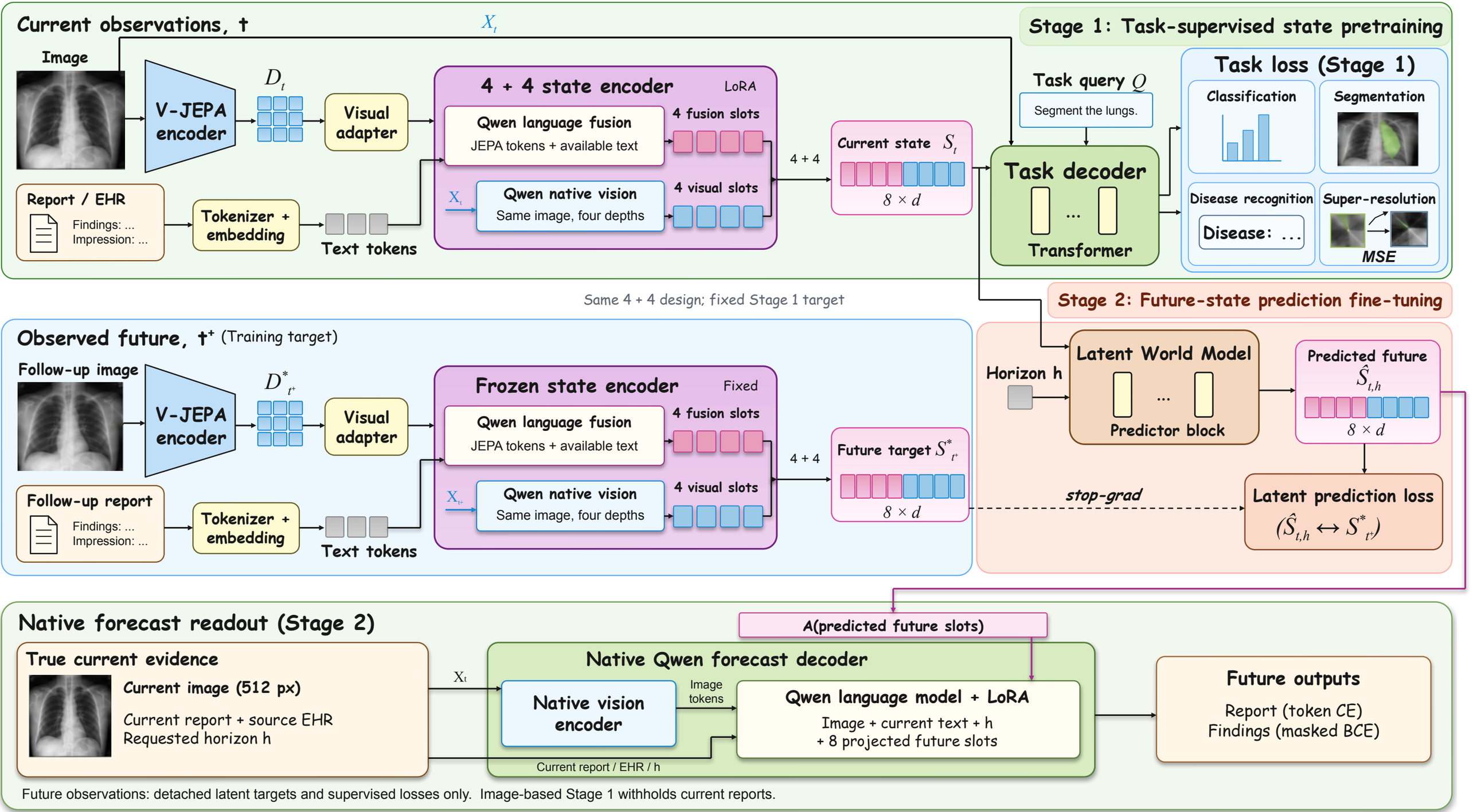

Current observations, t
X_t
Image
V-JEPA encoder
D_t
Visual adapter
4 + 4 state encoder
LoRA
Qwen language fusion
JEPA tokens + available text
4 fusion slots
Qwen native vision
Same image, four depths
4 visual slots
Report / EHR
Findings: ...
Impression: ...
Tokenizer + embedding
Text tokens
4 + 4
Current state S_t
$8 \times d$
Stage 1: Task-supervised state pretraining
Task query Q
Segment the lungs.
Task decoder
...
Transformer
Task loss (Stage 1)
Classification
Segmentation
Disease recognition
Disease: ...
Super-resolution
MSE
Same 4 + 4 design; fixed Stage 1 target
Stage 2: Future-state prediction fine-tuning
Observed future, t+ (Training target)
Follow-up image
V-JEPA encoder
$D^*_{t^+}$
Visual adapter
Frozen state encoder
Fixed
Qwen language fusion
JEPA tokens + available text
4 fusion slots
Qwen native vision
Same image, four depths
4 visual slots
Follow-up report
Findings: ...
Impression: ...
Tokenizer + embedding
Text tokens
4 + 4
Future target $S^*_{t^+}$
$8 \times d$
Horizon h
Latent World Model
...
Predictor block
Predicted future
$\hat{S}_{t,h}$
$8 \times d$
stop-grad
Latent prediction loss
$(\hat{S}_{t,h} \leftrightarrow S^*_{t^+})$
Native forecast readout (Stage 2)
A(predicted future slots)
True current evidence
Current image (512 px)
Current report + source EHR
Requested horizon h
X_t
Native Qwen forecast decoder
Native vision encoder
Image tokens
Qwen language model + LoRA
Image + current text + h
+ 8 projected future slots
Current report / EHR / h
Future outputs
Report (token CE)
Findings (masked BCE)
Future observations: detached latent targets and supervised losses only. Image-based Stage 1 withholds current reports.